

▼ **Figure 46.17** Some stages of human development during the first and second trimesters.



(a) **5 weeks.** Limb buds, eyes, the heart, the liver, and rudiments of all other organs have started to develop in the embryo, which is only about 1 cm long.



(b) **14 weeks.** Growth and development of the offspring, now called a fetus, continue during the second trimester. This fetus is about 6 cm long.

This disk-shaped organ contains both embryonic and maternal blood vessels. Exchange between the maternal and embryonic circulatory systems supplies nutrients, provides immune protection, exchanges respiratory gases, and disposes of metabolic wastes for the embryo. Blood from the embryo travels to the placenta through the arteries of the umbilical cord and returns via the umbilical vein.

Occasionally, an embryo splits during the first month of development, resulting in identical, or *monozygotic* (one-egg), twins. In contrast, fraternal, or *dizygotic*, twins arise when two follicles mature in a single cycle, followed by independent fertilization and implantation of two genetically distinct embryos.

The first trimester is the main period of **organogenesis**, the development of the body organs (**Figure 46.17a**). During organogenesis, the embryo is particularly susceptible to damage. For example, alcohol that passes through the placenta and reaches the developing central nervous system of the embryo can cause developmental and intellectual disabilities and other serious symptoms of fetal alcohol syndrome. The heart begins beating by the 4th week; a heartbeat can be detected at 8–10 weeks. At 8 weeks, all the major structures of the adult are present in rudimentary form, and the embryo is called a **fetus**.

At the end of the first trimester, the fetus is well differentiated, but only 5 cm long. Meanwhile, a high progesterone level brings about rapid changes in the mother: Mucus in the cervix forms a plug that protects against infection, the maternal part of the placenta grows, the mother's breasts and uterus get larger, and both ovulation and menstrual cycling stop. About three-fourths of all pregnant women experience

nausea, misleadingly called “morning sickness,” during the first trimester.

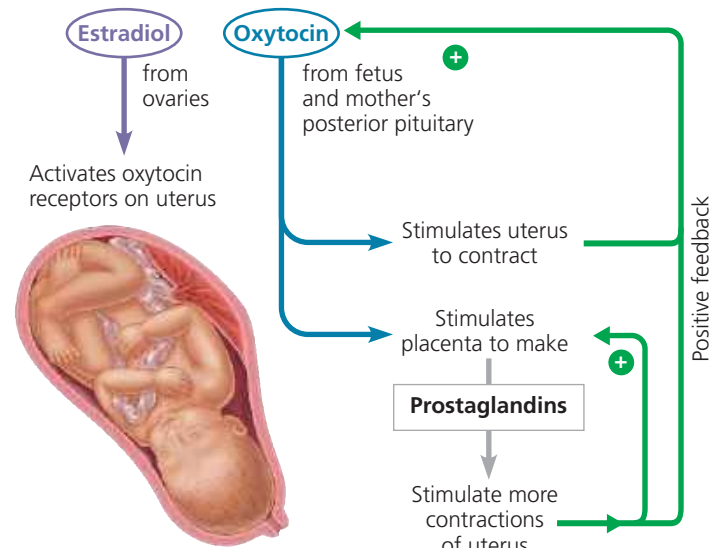
Second and Third Trimesters

During the second trimester, the fetus grows to about 30 cm in length. Development continues, including formation of fingernails, external sex organs, and outer ears (**Figure 46.17b**). The mother may feel fetal movements as early as one month into the second trimester, and fetal activity is typically visible through the abdominal wall one to two months later. Hormone concentrations stabilize as hCG secretion declines; the corpus luteum deteriorates; and the placenta completely takes over the production of progesterone, the hormone that maintains the pregnancy.

During the third trimester, the fetus grows to about 3–4 kg in weight and 50 cm in length. Fetal activity may decrease as the fetus fills the available space. As the fetus grows and the uterus expands around it, the mother's abdominal organs become compressed and displaced, leading to digestive blockages and a need for frequent urination.

Childbirth begins with **labor**, a series of strong, rhythmic uterine contractions that push the fetus and placenta out of the body. Once labor begins, local regulators (prostaglandins) and hormones (chiefly estradiol and oxytocin) induce and regulate further contractions of the uterus (**Figure 46.18**). Central

▼ **Figure 46.18** Positive feedback in labor.



VISUAL SKILLS Based on the feedback circuits shown, predict the effect of a single dose of oxytocin on a pregnant woman at the end of 39 weeks gestation.